

NEWPORT NEWS WILLIAMSBURG, VA, USA

WMO: 723086

Lat: 37.132N

Lon: 76.493W

Elev: 42

StdP: 14.67

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 93741

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	19.4	23.3	0.7	5.7	24.8	5.7	7.4	28.3	24.0	46.7	20.9	47.1	6.8	0	0.477

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.5	94.3	77.1	91.7	76.2	89.8	75.4	79.5	90.1	78.3	88.0	77.3	86.3	10.4	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
76.8	140.0	84.4	75.4	133.4	82.9	74.7	129.9	82.2	43.0	90.2	41.7	88.3	40.6	86.4	86.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 20.5	18.5	16.9	DB	12.5	99.1	5.7	3.5	8.3	101.7	5.0	103.7	1.8	105.7	-2.3	108.3	(4)
(5)			WB	10.5	82.3	5.1	2.0	6.8	83.8	3.9	85.0	1.0	86.1	-2.7	87.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	60.4	40.3	42.7	49.3	59.3	67.5	75.8	79.7	78.1	72.5	62.0	51.3	44.5		
	DBStd	16.01	10.73	9.53	9.47	8.80	7.69	6.28	4.97	4.67	5.84	7.88	8.58	9.49		
	HDD50	1048	339	243	133	16	0	0	0	0	0	5	85	227		
	HDD65	3389	767	624	493	216	65	5	0	0	9	155	417	638		
	CDD50	4827	38	40	112	296	543	775	922	871	674	376	124	56		
	CDD65	1691	1	1	6	46	143	329	457	406	233	61	6	2		
	CDH74	15722	2	6	61	410	1216	3193	4824	3870	1713	394	25	8		
	CDH80	5854	0	0	6	101	386	1274	2069	1456	481	80	1	0		
(14) Wind	WSAvg	8.0	8.3	8.5	9.3	9.3	8.3	7.8	7.6	7.2	7.4	7.3	7.5	7.6		
(15) Precipitation	PrecAvg	46.9	3.7	3.6	4.2	2.9	4.0	4.2	4.9	4.8	4.7	3.1	3.1	3.5		
	PrecMax	68.0	8.7	5.9	10.9	8.4	8.7	10.5	10.6	9.8	15.5	12.0	6.7	7.6		
	PrecMin	28.3	0.7	0.2	1.1	0.5	0.4	0.6	1.2	0.6	0.6	0.4	0.3	0.4		
	PrecStd	10.0	1.5	1.4	2.0	1.7	2.0	2.5	2.4	2.5	3.5	2.2	1.8	1.8		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	72.5	74.4	80.7	87.5	91.4	97.2	98.3	95.8	92.1	86.3	77.2	74.5		
		MCWB	63.2	63.0	63.8	68.0	73.6	76.9	78.1	78.1	74.7	71.5	65.9	64.6		
	2%	DB	67.4	68.5	74.8	82.9	88.0	92.9	94.8	92.6	88.6	82.0	73.2	69.4		
		MCWB	60.8	57.9	60.8	66.1	72.2	76.0	77.8	76.9	74.0	69.9	63.7	62.8		
	5%	DB	63.1	63.9	70.3	79.1	84.4	90.1	92.2	90.2	85.3	78.5	70.0	64.4		
		MCWB	56.9	56.0	58.5	64.5	70.6	75.1	76.9	75.9	73.0	68.3	62.5	59.3		
	10%	DB	57.4	59.2	65.9	74.6	81.1	87.4	89.8	87.7	82.2	75.0	65.8	60.4		
		MCWB	51.8	51.3	56.8	62.8	68.9	74.0	76.3	75.1	72.0	66.8	59.7	55.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	65.5	64.5	66.7	71.5	76.5	79.3	81.4	81.3	78.2	75.4	69.6	67.6		
		MCDB	70.8	71.1	75.5	81.6	86.6	90.7	92.2	91.8	86.4	79.5	73.3	72.5		
	2%	WB	61.7	61.4	63.9	68.9	74.2	77.8	80.0	79.3	76.8	73.1	66.3	63.6		
		MCDB	66.6	66.9	70.6	78.5	83.8	89.0	90.9	88.7	83.8	78.9	71.0	67.8		
	5%	WB	57.5	57.2	61.4	66.7	72.4	76.7	78.7	78.0	75.7	70.8	63.6	60.1		
		MCDB	61.8	62.8	69.0	75.8	81.4	86.9	88.9	86.8	81.8	76.7	67.9	64.1		
	10%	WB	52.5	52.2	57.9	64.3	70.7	75.5	77.6	76.9	74.3	68.1	60.4	55.7		
		MCDB	56.9	58.3	64.7	72.5	78.9	84.4	86.9	84.8	79.9	73.7	65.3	60.1		
(35) Mean Daily Temperature Range	5% DB	MDBR	17.3	18.9	19.6	20.8	19.1	18.0	17.5	16.9	17.1	19.1	19.4	17.7		
		MCDBR	23.4	25.4	25.6	24.6	22.0	20.2	19.7	18.8	19.5	21.3	22.5	21.2		
		MCWBR	17.8	17.9	15.4	11.9	10.5	8.1	7.4	7.4	8.4	11.5	14.5	16.5		
	5% WB	MCDBR	21.4	23.8	22.3	21.8	19.4	18.0	17.9	16.5	15.3	16.6	18.8	19.7		
		MCWBR	18.3	19.1	16.6	12.3	10.1	7.9	8.0	7.3	7.7	10.0	14.4	17.0		
(40) Clear-Sky Solar Irradiance	taub		0.306	0.318	0.345	0.392	0.434	0.485	0.514	0.509	0.409	0.353	0.322	0.310		
	taud		2.541	2.502	2.456	2.337	2.249	2.156	2.090	2.091	2.361	2.513	2.551	2.562		
	Ebn at Noon		282	292	292	282	271	256	247	245	267	275	274	273		
	Edh at Noon		24	29	33	39	44	48	51	50	36	28	24	23		
(44) All-Sky Solar Radiation	RadAvg		742	981	1305	1657	1870	2014	1918	1730	1404	1114	839	659		
	RadStd		60	107	105	121	167	134	94	92	120	130	69	52		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	+0.90	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+254	+178
Regional (68 neighbors)	+0.83	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+290	+187

Nomenclature: See separate page